

Restaurant Market Analysis Report

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Part 1: Data Collection

A dataset of 30 restaurants was collected from the College Road locality in Nashik using Zomato. The objective was to understand the local food delivery market and identify business opportunities.

The following attributes were collected for each restaurant:

- Restaurant Name
- Cuisine
- Rating
- Number of Reviews
- Cost for Two
- Delivery Time
- Restaurant Type

Restaurant types were categorized as Dine-In Restaurant, Cloud Kitchen, or Unknown based on available information.

The final dataset contained 30 restaurants from different cuisine categories including North Indian, South Indian, Fast Food, Pizza, Biryani, Chinese, Healthy Food, and Street Food.

Part 2: Network Investigation

Chrome Developer Tools were used to investigate the network activity of the food delivery platform.

The Network tab was opened and Fetch/XHR requests were inspected. Multiple requests related to application initialization, activity tracking, and platform communication were observed.

The following observations were made:

- Network requests primarily returned JSON responses.
- Requests contained platform configuration and communication data.
- Fetch/XHR requests were used for data exchange between the client and server.
- Request headers, status codes, and response structures were reviewed.

Screenshots of the investigation process have been included as supporting evidence.

Part 3: Menu Intelligence

Five restaurants were selected for menu analysis:

- Subway
- Faasos – Wraps, Rolls & Shawarma
- Pizza Hut
- Hakka Tikka
- Burger King

For each restaurant, ten menu items were collected along with their categories and prices.

Key observations:

- Pizza Hut had the highest priced menu items among the selected restaurants.
- Burger King offered the most affordable entry-level items.
- Subway showed a strong focus on sandwiches and salads.
- Faasos specialized in wraps and shawarma products.
- Hakka Tikka offered a diverse menu including noodles, rice, biryani, and chicken dishes.

The collected menu data was used to identify pricing patterns, menu diversity, and potential bestseller products.

PART 4: DATA CLEANING

Objective

The collected restaurant dataset was reviewed and cleaned to improve data quality, consistency, and reliability before performing further analysis.

Cleaning Process

1. Duplicate Detection

The dataset was checked for duplicate restaurant records using restaurant names and related attributes. No duplicate records were found in the dataset.

2. Missing Value Handling

The dataset was reviewed for missing values in key columns such as Restaurant Name, Cuisine, Rating, Cost for Two, and Delivery Time. No critical missing values were identified. Restaurants with unclear classifications were marked as "Unknown" in the Type column.

3. Data Standardization

Cuisine names, restaurant types, and delivery time formats were reviewed for consistency. Restaurant types were standardized into three categories:

- Dine-In Restaurant
- Cloud Kitchen
- Unknown

4. Data Validation

Ratings, review counts, costs, and delivery times were cross-verified with the source platform to ensure accuracy.

Cleaning Outcome

Total Records Before Cleaning: 30

Total Records After Cleaning: 30

Duplicate Records Removed: 0

Missing Values Handled: 0

The dataset was found to be clean and ready for business analysis.

Part 5: Business Analysis

1. Most Saturated Cuisine

Based on the collected dataset, North Indian cuisine was found to be the most saturated cuisine in the selected locality. Multiple restaurants offered North Indian dishes either as their primary cuisine or as part of a mixed cuisine menu. This indicates strong customer demand but also high competition within this segment.

2. Underrepresented Cuisine

Gujarati and Kathiyawadi cuisines were found to be underrepresented in the locality. Only a limited number of restaurants offered these cuisines, suggesting a potential opportunity for new entrants targeting this niche market.

3. Cloud Kitchen Recommendation (Budget: ₹5 Lakh)

If launching a cloud kitchen with a budget of ₹5 lakh, I would recommend focusing on Healthy Bowls and North Indian Meals.

Target Audience:

- ❖ Students
- ❖ Working Professionals
- ❖ Health-conscious customers

Pricing Strategy:

- ❖ Average Order Value: ₹200–₹350

Reasons:

- ❖ Growing demand for convenient and healthy meal options
 - ❖ Lower operational costs compared to dine-in restaurants
 - ❖ Strong delivery compatibility
4. Most Operationally Efficient Restaurant

Most Operationally Efficient Restaurant

Based on the collected dataset, Larive Kitchen and Cocktail appears to be one of the most operationally efficient restaurants.

Reasons:

- ❖ Highest rating (14.6K)
- ❖ Very high review count (4.5)
- ❖ Consistent delivery performance (30–35 mins)
- ❖ Strong customer engagement and popularity

These factors indicate strong operational consistency and customer satisfaction.

Part 6: SQL Analysis

-- Database Schema

```
CREATE TABLE Restaurants (
  restaurant_id INT PRIMARY KEY,
  restaurant_name VARCHAR(100),
  cuisine VARCHAR(255),
  rating DECIMAL(2,1),
  reviews VARCHAR(20),
  cost_for_two INT,
```

```
locality VARCHAR(100),
delivery_time VARCHAR(20),
restaurant_type VARCHAR(50)
);

CREATE TABLE Menu_Items (
    item_id INT PRIMARY KEY,
    restaurant_id INT,
    menu_category VARCHAR(100),
    item_name VARCHAR(255),
    price DECIMAL(10,2),
    FOREIGN KEY (restaurant_id)
    REFERENCES Restaurants(restaurant_id)
);
```

– SQL QUERIES

Query 1: Top 5 Highest Rated Restaurants

Answer - Select restaurant_name, rating
From Restaurants
Order By rating DESC
Limit 5;

Query 2: Average Cost for Two by Cuisine

Answer - Select cuisine,
AVG(cost_for_two) as average_cost
From restaurants
Group by cuisine;

Query 3: Restaurants with More Than One Cuisine Tag

Answer - Select restaurant_name, cuisine
From Restaurants
Where cuisine LIKE '%,%';

Query 4: Highest Priced Menu Item

Answer - Select item_name, price
From Menu_Items
Order By price DESC
Limit 1;

Part 7: Methodology Report

1. How Data Was Collected

For this project, I selected the College Road locality in Nashik and collected restaurant information from Zomato. I manually recorded details such as restaurant name, cuisine, rating, review count, cost for two, delivery time, and restaurant type in Google Sheets. A total of 30 restaurants were included in the dataset.

To complete the menu analysis section, I selected five restaurants from the collected data and noted down their menu categories, item names, and prices. The information was organized in a structured format for further analysis.

2. Tools Used

The following tools were used during the project:

- ❖ Google Sheets for storing and organizing data
- ❖ Google Chrome for browsing restaurant listings
- ❖ Chrome Developer Tools for viewing network requests
- ❖ Google Docs for preparing the report
- ❖ SQL for designing database schema and queries

3. Challenges Faced

While working on the project, I faced a few challenges. In some cases, it was not easy to determine whether a restaurant should be classified as a cloud kitchen or a dine-in restaurant. Some restaurants also had multiple cuisine tags, which made categorization slightly difficult. During the network investigation task, many requests were displayed, so identifying the relevant ones required additional observation.

4. Assumptions

A few assumptions were made during the analysis:

- ❖ Restaurant information displayed on the platform was considered accurate.
- ❖ Delivery times were recorded as shown during data collection.
- ❖ Cuisine categories were taken directly from the restaurant listings.
- ❖ Restaurants focused mainly on delivery were categorized as cloud kitchens where appropriate.

5. Improvements for Future Work

If I had more time, I would collect data from additional localities to make the analysis broader. I would also include customer review analysis and compare pricing patterns across different cuisine categories. This would help generate more detailed business insights.

